EXPERIMENT – 4

AIM :

To write a Python program to solve a Crypt-Arithmetic Problem by assigning unique digits to letters to satisfy a mathematical equation.

ALGORITHM:

1. Start.
2. Read the given arithmetic equation containing letters.
3. Identify all unique letters in the equation.
4. Assign different digits from 0 to 9 to the letters.
5. Ensure that the first letter of each number is not assigned zero.
6. Evaluate the equation using the assigned digits.
7. Check whether the equation is mathematically correct.
8. Display the valid digit assignment.
9. Stop.

PROGRAM :

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```
from itertools import permutations

letters = "SENDMORY"

for p in permutations(range(10), 8):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
        continue

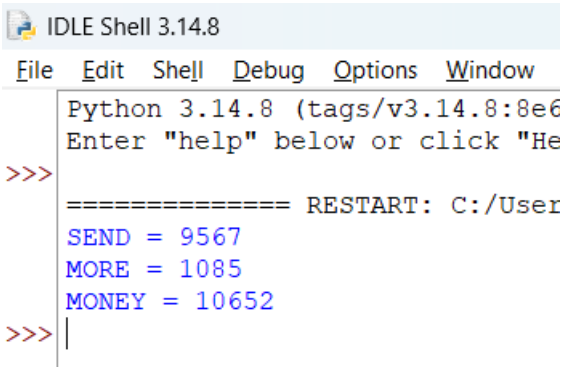
    SEND = (d['S']*1000 + d['E']*100 +
            d['N']*10 + d['D'])

    MORE = (d['M']*1000 + d['O']*100 +
            d['R']*10 + d['E'])

    MONEY = (d['M']*10000 + d['O']*1000 +
             d['N']*100 + d['E']*10 + d['Y'])

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

OUTPUT:



RESULT:

Thus, the Python program to solve the Crypt-Arithmetic Problem was executed successfully, and a valid digit assignment satisfying the equation was obtained